Interior Point Method - Proof Outline

The basic formulation

$F_0(x)$ – barrier function: maintains feasibility

$\min: c^T x$

$\text{st}: x \in \mathcal{X}$

$\min: F_t(x) = t \cdot c^T x + F_0(x)$

Letting $t \to \infty$

Three key objectives

✓ 1. Minimize F_t quickly if we start close enough to $x_t \triangleq \operatorname{argmin}: F_t(x)$. We showed: $\lambda_F(x) \leq 1/4$ $\Rightarrow$ quad. convergence.

2. Increase t quickly: $t^+ = (1+\alpha)\, t$

3. $c^T x_t - c^T x^* \leq O(1/t)$

Step 2: Increasing t

Recall flow of IPM:

t_0 initialize at 0.

$t^+ = (1+\alpha)\, t$

Need: $\lambda_{F_{t^+}}(x_t) \leq 1/4$.

Step 2: Increasing t — F_t and F_0

Need: $\boxed{\lambda_{F_{t^+}}(x_t) \leq 1/4.}$

$$\boxed{x_t = \operatorname{argmin}_x F_t(x)} = \operatorname{argmin}_x \underbrace{t\,c^T x + F_0(x)}_{F_t}$$

Recall: $\lambda_F(x) \triangleq \|\nabla F(x)\|_x^*$

Note: $F_{t_1}(x) = (t_1 - t_2)\,c^T x + F_{t_2}(x)$

use of this local norm is critical.

$$\lambda_{F_{t^+}}(x_t) = \|\nabla F_{t^+}(x_t)\|_{x_t}^*$$

$$= \|t^+ c + \nabla F_0(x_t)\|_{x_t}^* = \|(t^+ - t)\,c + \boxed{\cancel{t\,c + \nabla F_0(x_t)}}\|_{x_t}^*$$

$\nabla F_t(x_t) \to 0$

$$= \|(t^+ - t)\cdot c\|_{x_t}^* = (t^+ - t)\,\|c\|_{x_t}^*$$

$$= \alpha \cdot t \cdot \|c\|_{x_t}^* \leq 1/4$$

want

Thus: $\|c\|_{x_t}^* \to O(1/t)$.

Step 2: Increasing t — local norm is critical

Summary:

$$\|c\|^*_{x_t} \sim O(1/t)$$

$$\|c\|^*_{x_t} = \sup_{\|h\|_{x_t}} : h^T c = \sup_{h^T \underbrace{\nabla^2 F_0(x_t)}\, h \le 1} : h^T c$$

(exercise)
$$= \sqrt{c^T \left(\nabla^2 F_0(x_t)\right)^{-1} c}$$

Intuition: $\nabla^2 F_0(x_t)$ - small as $x_t \to$ bdry.

Step 2: Increasing t — local norm is critical — continued

An Example

$\min : c^T x$

$\text{st}: x \geq 0$

$F_t(x) = tcx - \log x$

$F_0(x) = -\log x$

$\nabla F_t(x) = tc - \frac{1}{x}$

$\nabla^2 F_t(x) = \underline{\nabla^2 F_0(x) = \frac{1}{x^2}}$

$x_t = \text{argmin}: t \cdot cx - \log x$

Take deriv, set $= 0$

$tc - \frac{1}{x} = 0$

$$\boxed{x_t = \frac{1}{tc}}$$

$\|c\|^*_{x_t} = \sup : c \cdot h$

$\text{st}: \underline{\|h\|^2_{x_t}} \leq 1^2$

$= \sup : ch$

$\text{st}: \frac{h^2}{x_t^2} \leq 1$

$= |c \cdot x_t|$

$= \frac{1}{t}$

$$\boxed{t^+ = t + \frac{1}{4\|c\|^*_{x_t}} = t + \frac{1}{4 \cdot \frac{1}{t}}}$$

α

$= (1 + \frac{1}{4})t$

We wanted:

$\lambda_{F_{t^+}}(x_t) = \alpha \cdot t \cdot \|c\|^*_{x_t} = \alpha$

$\Rightarrow$ Take $\alpha = 1/4$

Increasing t — an additional condition —

Recap: Need $\lambda_{F_{t^+}}(x_t) \leq 1/4$

and: $\lambda_{F_{t^+}}(x_t) = \underline{(t^+ - t) \cdot \|c\|^*_{x_t}} \leq 1/4$

To satisfy: we can take t^+ as large as

$$t^+ = t + \frac{1}{4\|c\|^*_{x_t}}$$

Using $\nabla F_t(x_t) = 0 = tc + \nabla F_0(x_t)$

$$\Rightarrow c = -\frac{1}{t} \nabla F_0(x_t)$$

$$\underline{\|c\|^*_{x_t}} = \underline{\frac{1}{t}} \, \underline{\|\nabla F_0(x_t)\|^*_{x_t}}$$

For $\|c\|^*_{x_t} \sim \frac{1}{t}$ need: $\|\nabla F_0(x_t)\|^*_{x_t}$

$\sim O(1)$

$F_0(x) = -\sum \log(b_i - a_i^T x)$ satisfies this.

Recall:

$$\|\nabla F_0(x_t)\|^*_{x_t} = \sqrt{\nabla F_0(x_t)^T (\nabla^2 F_0(x_t))^{-1} \nabla F_0(x_t)}$$

We will essentially add a condition that guarantees this.

Increasing t — an additional condition — self concordant barrier

Self-Concordant Barrier

A M-self-concordant function F is called a ν-self-concordant barrier, if:

$$\nabla^2 F(x) \succeq \frac{1}{\nu}\left(\nabla F(x) \nabla F(x)^T\right)$$

Increasing t — an additional condition — self concordant barrier

If we have this property: $\nabla^2 F(x) \succeq \frac{1}{\nu} \nabla F(x) \nabla F(x)^T$

$$\|\nabla F(x)\|_x^* = \sup : \nabla F(x)^T h$$

$$\text{st}: \; h^T \left(\underline{\nabla^2 F(x)}\right) h \leq 1$$

$$\leq \sup : \nabla F(x)^T h$$

$$h^T \left(\nabla F(x) \nabla F(x)^T \right) \leq \nu$$

$$= \sqrt{\nu}$$

Increasing t — which functions are self concordant barriers?

Lemma: If $G(x) = \exp\left\{-\frac{1}{\nu} F_0(x)\right\}$ is concave

then: $\nabla^2 F_0(x) \succeq \frac{1}{\nu} \nabla F_0(x) \nabla F_0(x)^T$.

Ex: $-\log x$

$-\log\det X$

Are both self-concordant barriers.

Step 2: Increasing t — putting it all together

1. $t^+ = t + \dfrac{1}{4\|c\|^*_{x_t}}$

2. $\|c\|^*_{x_t} = \frac{1}{t} \|\nabla F_0(x_t)\|^*_{x_t}$

3. If F_0 is a ν - self-conc. barrier

$\Rightarrow \|\nabla F_0(x_t)\|^*_{x_t} \leq \sqrt{\nu}$

$$\Rightarrow t^+ = \underbrace{\left(1 + \frac{1}{4\sqrt{\nu}}\right)} \cdot t$$

Can increase t at a linear rate.

Step 3: convergence to optimal solution

Fact:

$$c^T x_t - c^T x^*_{LP} \leq \nu/t .$$

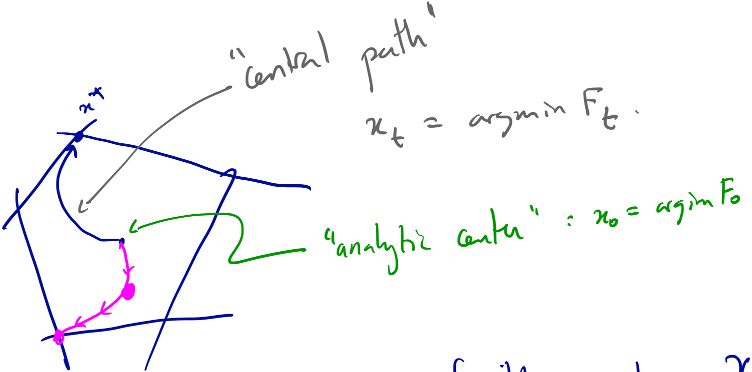
sol'n to LP

Need: $\log(1/\varepsilon)$ iterations for ε-suboptimal solution.

Initializing the Central Path IPM

min: $c^T x$

st: $x \in$



We can get to central path using any feasible point $y \in \mathcal{X}$.

$$y \in \mathcal{X} \Rightarrow y \in \operatorname{argmin}: -t\left(\nabla F_0(y)\right)^T x + F_0(x) \quad \text{at } t = 1$$

Instead of increasing t, we decrease from 1 to 0 at a geometric rate.